

Answer: A

10 Using $pV = nRT$

$$500000 \times 4.0 \times 10^{-4} = n_1 R(300)$$

$$n_1 = 0.0802 \text{ moles}$$

After filling the cylinder,

$$620000 \times 4.0 \times 10^{-4} = n_2 R(300)$$

$$n_2 = 0.0995 \text{ moles}$$

Amount pumped in = 0.019 moles

Answer: A

11 Using 1st law of thermodynamics